BEAM DESIGN SUMMARY

Project Name	:	G+2 RB BEAM DESIGN @MAIN ROOF
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-11.125 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	11.125 m

Beam No		:	**B1**	
Group No		:	**G1**	
Analysis Reference(Member)	11.125 m	:	7	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	27.42	25.88	11.6	25.09	9.59	22.99
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	25.76	21.68	31.49
Asv Torsion (sqmm/m)	93.44	88.94	120.01
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B2**	
Group No		:	**G1**	
Analysis Reference(Member)	11.125 m	:	8	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	30.62	6.34	3.13	19.05	65.24	111.19
PtClc (%)	0.213	0.2	0.2	0.2	0.485	0.903
AstCalc (sqmm)	181.49	170.2	170.2	170.2	412.78	768.43
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	33.65	43.47	53.28
Asv Torsion (sqmm/m)	118.7	149.76	180.83
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B3**	
Group No		:	**G1**	
Analysis Reference(Member)	11.125 m	:	142	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm

Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.44	16.44	21.52	101.44	76.04	30.52
PtClc (%)	0.2	0.2	0.2	0.823	0.579	0.213
AstCalc (sqmm)	170.2	170.2	170.2	700.45	492.5	180.9
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	117.49	112.68	104.94
Asv Torsion (sqmm/m)	556.67	541.44	516.96
Asv Reqd (sqmm/m)	455.16	419.15	361.25
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		**:**	**B4**	
Group No		**:**	**G2**	
Analysis Reference(Member)	11.125 m	:	27	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	33.28	21.31	18.4	22.65	12.56	18.09
PtClc (%)	0.233	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	198.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48

Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	13.74	15.73	18.44
Asv Torsion (sqmm/m)	-	52.94	61.52
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B5**	
Group No		:	**G2**	
Analysis Reference(Member)	11.125 m	:	28	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	40.3	5.35	2.15	13.8	70.44	106.74
PtClc (%)	0.285	0.2	0.2	0.2	0.529	0.868
AstCalc (sqmm)	242.95	170.2	170.2	170.2	450.59	738.41
Ast Prv (sqmm)	339.3	226.2	226.2	942.48	942.48	942.48
Reinforcement	3-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	40.78	40.78	40.64
Asv Torsion (sqmm/m)	145.71	145.71	140.83
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B6**	
Group No		:	**G2**	
Analysis Reference(Member)	11.125 m	:	147	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	13.97	13.97	26.14	106.53	78.5	25.97
PtClc (%)	0.2	0.2	0.2	0.866	0.601	0.2
AstCalc (sqmm)	170.2	170.2	170.2	737.01	511.33	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	117.64	115.84	108.53
Asv Torsion (sqmm/m)	513.93	508.25	485.09
Asv Reqd (sqmm/m)	456.27	442.84	573.79
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B7**	
Group No		:	**G3**	
Analysis Reference(Member)	11.125 m	:	47	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm

Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	31.45	18.34	19.21	23.37	13.01	24.64
PtClc (%)	0.219	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	186.72	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	16.77	18.51	21.22
Asv Torsion (sqmm/m)	66.17	67.37	75.85
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B8**	
Group No		**:**	**G3**	
Analysis Reference(Member)	11.125 m	:	48	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	30.23	7.85	7.73	15.82	40.45	63.4
PtClc (%)	0.21	0.2	0.2	0.2	0.287	0.47

AstCalc (sqmm)	179.05	170.2	170.2	170.2	243.92	399.58
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	27.15	27.15	29.89
Asv Torsion (sqmm/m)	100.64	100.64	108.15
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B9**	
Group No		:	**G3**	
Analysis Reference(Member)	11.125 m	:	152	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	14.88	14.88	14.88	64.43	50.94	28.9
PtClc (%)	0.2	0.2	0.2	0.478	0.368	0.201
AstCalc (sqmm)	170.2	170.2	170.2	406.94	313.18	170.8
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	73.55	73.14	71.85
Asv Torsion (sqmm/m)	370.67	369.38	365.34
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14

Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B10**	
Group No		:	**G4**	
Analysis Reference(Member)	11.125 m	:	67	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	31.85	20.2	23.53	31.45	15.36	34.25
PtClc (%)	0.222	0.2	0.2	0.219	0.2	0.24
AstCalc (sqmm)	189.2	170.2	170.2	186.68	170.2	204.33
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	23.88	22.61	28.93
Asv Torsion (sqmm/m)	83.51	79.18	98.94
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B11**	
Group No		:	**G4**	
Analysis Reference(Member)	11.125 m	:	68	
Breadth		:	230	mm

Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	34.66	14.55	7.63	21.14	36.53	64.67
PtClc (%)	0.243	0.2	0.2	0.2	0.257	0.48
AstCalc (sqmm)	206.89	170.2	170.2	170.2	218.76	408.7
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	28.32	33.61	39.93
Asv Torsion (sqmm/m)	112.93	132.08	151.84
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B12**	
Group No		**:**	**G4**	
Analysis Reference(Member)	11.125 m	:	157	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom	Beam Top

	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	8.54	8.54	17.07	67.08	48.1	15.66
PtClc (%)	0.2	0.2	0.2	0.501	0.346	0.2
AstCalc (sqmm)	170.2	170.2	170.2	426.07	294.14	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	68.07	67.18	64.38
Asv Torsion (sqmm/m)	265.24	262.43	253.68
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B13**	
Group No		:	**G5**	
Analysis Reference(Member)	11.125 m	:	132	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	12.7	17.68	8.47	13.34	2.18	18.52
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	33.35	17.33	36.93

Asv Torsion (sqmm/m)	138.6	89.04	149.68
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B14**	
Group No		:	**G5**	
Analysis Reference(Member)	11.125 m	:	133	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	8.31	11.88	4	13.5	5.25	24.78
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	26.39	8.27	31.9
Asv Torsion (sqmm/m)	93.47	-	110.51
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No	:	**B15**
Group No	:	**G5**

Analysis Reference(Member)	11.125 m	:	160	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	17.87	17.87	17.87	34.33	30.79	27.69
PtClc (%)	0.2	0.2	0.2	0.241	0.215	0.2
AstCalc (sqmm)	170.2	170.2	170.2	204.78	182.57	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	85.41	81.29	75.74
Asv Torsion (sqmm/m)	554.86	542.11	524.94
Asv Reqd (sqmm/m)	400.74	369.89	328.35
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		**:**	**B16**	
Group No		**:**	**G6**	
Analysis Reference(Member)	11.125 m	:	129	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	

Beam Type : Cantilever Beam

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	21.1	6.94	6.94	9.35	55.87	85.67
PtClc (%)	0.2	0.2	0.2	0.2	0.408	0.667
AstCalc (sqmm)	170.2	170.2	170.2	170.2	346.8	567.61
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	54.03	64.57	71.58
Asv Torsion (sqmm/m)	224.75	257.74	279.67
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B17**	
Group No		:	**G6**	
Analysis Reference(Member)	11.125 m	:	114	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	4.21	14.34	27.3	81.4	40.01	20.85
PtClc (%)	0.2	0.2	0.2	0.627	0.283	0.2
AstCalc (sqmm)	170.2	170.2	170.2	533.77	241.08	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	41.05	29.81	18.56
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B18**	
Group No		:	**G6**	
Analysis Reference(Member)	11.125 m	:	102	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	9.72	15.25	15.25	37.47	8.36	28.34
PtClc (%)	0.2	0.2	0.2	0.264	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	224.76	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	30.71	17.6	25.7
Asv Torsion (sqmm/m)	105.41	63.55	86.82
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B19**	
Group No		:	**G6**	
Analysis Reference(Member)	11.125 m	:	90	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	8.67	21.28	21.82	44.47	11.69	27.65
PtClc (%)	0.2	0.2	0.2	0.317	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	270.11	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	34.26	20.4	27.11
Asv Torsion (sqmm/m)	118.98	84.89	101.17
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B20**	
Group No		:	**G7**	
Analysis Reference(Member)	11.125 m	:	130	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm

Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	27	3.36	3.36	5.54	65.96	102.32
PtClc (%)	0.2	0.2	0.2	0.2	0.491	0.832
AstCalc (sqmm)	170.2	170.2	170.2	170.2	417.97	708.28
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	55.15	66.76	70.1
Asv Torsion (sqmm/m)	179.92	216.68	227.24
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	718.14	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		**:**	**B21**	
Group No		**:**	**G7**	
Analysis Reference(Member)	11.125 m	:	115	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	14.04	18.99	50.7	79.77	44.98	19.99
PtClc (%)	0.2	0.2	0.366	0.612	0.321	0.2
AstCalc (sqmm)	170.2	170.2	311.58	521.13	273.52	170.2
Ast Prv (sqmm)	226.2	226.2	339.3	942.48	942.48	942.48

Reinforcement	2-T12	2-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	36.15	32.25	29.34
Asv Torsion (sqmm/m)	120.08	107.74	97.82
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B22**	
Group No		:	**G7**	
Analysis Reference(Member)	11.125 m	:	103	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	20.19	15	30.14	39.16	16.41	21.28
PtClc (%)	0.2	0.2	0.21	0.277	0.2	0.2
AstCalc (sqmm)	170.2	170.2	178.55	235.62	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	23.58	18.27	17.37
Asv Torsion (sqmm/m)	83.5	66.7	65.5
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B23**	
Group No		:	**G7**	
Analysis Reference(Member)	11.125 m	:	91	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.97	19.35	35.45	39.05	14.66	20.02
PtClc (%)	0.2	0.2	0.249	0.276	0.2	0.2
AstCalc (sqmm)	170.2	170.2	211.9	234.91	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	23.05	17.74	13.8
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B24**	
Group No		:	**G8**	
Analysis Reference(Member)	11.125 m	:	131	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm

Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	22.26	12.37	12.37	23.04	74.95	104.79
PtClc (%)	0.2	0.2	0.2	0.2	0.569	0.852
AstCalc (sqmm)	170.2	170.2	170.2	170.2	484.24	725.25
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	90.12	98.98	102.55
Asv Torsion (sqmm/m)	488.02	516.09	527.37
Asv Reqd (sqmm/m)	254.81	316.64	343.32
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		**:**	**B25**	
Group No		**:**	**G8**	
Analysis Reference(Member)	11.125 m	:	116	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	12.33	15.04	40.12	62.54	36.91	18.54
PtClc (%)	0.2	0.2	0.284	0.462	0.26	0.2
AstCalc (sqmm)	170.2	170.2	241.83	393.47	221.16	170.2

Ast Prv (sqmm)	226.2	226.2	339.3	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	29.51	25.96	25.96
Asv Torsion (sqmm/m)	105.12	98.23	98.23
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B26**	
Group No		:	**G8**	
Analysis Reference(Member)	11.125 m	:	104	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	24.72	20.5	35.7	37.65	18.81	26.39
PtClc (%)	0.2	0.2	0.251	0.265	0.2	0.2
AstCalc (sqmm)	170.2	170.2	213.52	225.94	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	34.31	30	28.34
Asv Torsion (sqmm/m)	177.37	163.71	153.34
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B27**	
Group No		:	**G8**	
Analysis Reference(Member)	11.125 m	:	92	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.59	25.44	25.44	52.69	12.72	43.57
PtClc (%)	0.2	0.2	0.2	0.382	0.2	0.311
AstCalc (sqmm)	170.2	170.2	170.2	325.03	170.2	264.24
Ast Prv (sqmm)	226.2	226.2	226.2	942.48	942.48	942.48
Reinforcement	2-T12	2-T12	2-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	43.6	28.44	35.96
Asv Torsion (sqmm/m)	163.78	115.79	130.77
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B28**	
Group No		:	**G9**	
Analysis Reference(Member)	11.125 m	:	161	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm

Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.58	16.58	16.58	29.53	44.26	56.63
PtClc (%)	0.2	0.2	0.2	0.205	0.316	0.414
AstCalc (sqmm)	170.2	170.2	170.2	174.72	268.76	352.04
Ast Prv (sqmm)	226.2	226.2	226.2	402.12	402.12	402.12
Reinforcement	2-T12	2-T12	2-T12	2-T16	2-T16	2-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	77.62	86.62	93.85
Asv Torsion (sqmm/m)	528.03	556.19	578.83
Asv Reqd (sqmm/m)	276.76	344.11	398.26
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		**:**	**B29**	
Group No		**:**	**G9**	
Analysis Reference(Member)	11.125 m	:	162	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	2.6	2.6	2.6	43.2	21.3	18.29

PtClc (%)	0.2	0.2	0.2	0.308	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	261.84	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	402.12	402.12	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T16	2-T16	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	31.02	17.99	22.87
Asv Torsion (sqmm/m)	120.37	79.59	102.02
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B30**	
Group No		:	**G9**	
Analysis Reference(Member)	11.125 m	:	163	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	8.5	16.95	12.96	28.8	8.54	25.8
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	43.12	43.12	47.62
Asv Torsion (sqmm/m)	271.64	271.64	251.59
Asv Reqd (sqmm/m)	254.81	254.81	254.81

Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B31**	
Group No		:	**G9**	
Analysis Reference(Member)	11.125 m	:	164	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	6.38	34.03	16.13	37.22	2.39	30.7
PtClc (%)	0.2	0.238	0.2	0.262	0.2	0.214
AstCalc (sqmm)	170.2	202.9	170.2	223.14	170.2	182.01
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	54.94	29.13	50.82
Asv Torsion (sqmm/m)	204.24	129.02	191.5
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -
